# ASSESSMENT TOOL - 1

## 1. Understand the Problem Statement

### Problem in Your Own Words

Blood banks are responsible for collecting, storing, and supplying blood to hospitals and patients. Managing blood inventory manually or with outdated systems often results in blood shortages, expired blood units, delayed donor identification, and inefficient emergency response. Since blood has a limited shelf life, poor inventory management leads to wastage while some hospitals face shortages during emergencies.

An intelligent blood bank management platform can use digital technologies such as Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and analytics to automate inventory management, predict future blood demand, monitor blood expiration, identify eligible nearby donors, and generate low-stock alerts. This improves blood availability and reduces wastage.

### Objectives of the Proposed System

The proposed system aims to:

- Digitally manage blood inventory.
- Monitor blood collection and expiration dates.
- Quickly identify compatible and eligible donors.
- Predict future blood demand using AI.
- Generate automatic alerts for low blood stock.
- Maintain secure donor records.
- Improve emergency blood availability.
- Reduce blood wastage through efficient inventory management.

### Scope of the Problem

The proposed system covers:

- Blood bank inventory management
- Donor registration and eligibility verification
- Blood request management
- Blood expiration tracking
- AI-based demand prediction
- Emergency donor identification

- Notification and alert system
- Hospital integration
- Cloud-based database management

The system does not include blood testing, laboratory diagnosis, or medical treatment processes.

## 2. Identify Key Stakeholders and Challenges

### Stakeholders

| Stakeholder | Role |
|---|---|
| Blood Bank Administrator | Manages inventory, donor records, and blood distribution |
| Donors | Register, donate blood, receive notifications |
| Hospitals | Request blood units during emergencies |
| Patients | Receive blood transfusions |
| Medical Staff | Verify donor eligibility and blood quality |
| Government Health Authorities | Monitor blood availability and compliance |
| System Administrator | Maintains software, database, and security |

### Roles and Responsibilities

**Blood Bank Administrator**

- Update blood inventory
- Monitor blood expiration
- Approve donor registrations
- Handle emergency blood requests

**Donors**

- Register personal information
- Update medical records
- Donate blood
- Respond to emergency requests

**Hospitals**

- Submit blood requests
- Receive available blood units
- Update emergency requirements

**Medical Staff**

- Verify donor eligibility
- Conduct blood screening
- Update medical records

**Government Authorities**

- Ensure regulatory compliance
- Monitor blood availability
- Generate healthcare reports

**System Administrator**

- Maintain database
- Manage user authentication
- Ensure cybersecurity
- Perform software updates

**Challenges Faced by Stakeholders**

| **Stakeholder** | **Challenges** |
|---|---|
| Blood Banks | Inventory shortage, expired blood units |
| Donors | Lack of reminders and awareness |
| Hospitals | Delayed blood availability |
| Patients | Emergency delays |
| Medical Staff | Manual record management |
| Government | Lack of real-time monitoring |

**3. Analyse the Current Approach**

**Existing System**

Many blood banks still use manual registers or standalone software systems for inventory management. Blood requests are usually handled through phone calls or manual verification. Donor information is often outdated, making emergency donor identification slow and inefficient.

**Strengths of Current System**

- Simple implementation
- Low initial cost
- Basic donor record management
- Familiar workflow for staff

**Limitations**

- Manual inventory updates
- Delayed donor identification
- No AI demand prediction
- No automatic expiration alerts
- Limited emergency response
- Higher blood wastage
- Lack of real-time monitoring

**Problems to be Addressed**

- Blood shortages
- Expired blood units
- Poor inventory visibility
- Delayed emergency response
- Lack of donor engagement
- Inaccurate demand estimation

## 4. Proposed Software Solution

**System Design**

The proposed solution is a cloud-based intelligent blood bank management platform integrated with AI and IoT technologies.

## Architecture

Donor Mobile App

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Hospital Portal

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Blood Bank Portal

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Cloud Database

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AI Prediction Engine

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Inventory Management

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IoT Blood Storage Monitoring

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Alert & Notification System

**Functional Modules**

**1. Donor Management**

- Registration
- Eligibility verification
- Donation history
- GPS location
- Notification system

**2. Blood Inventory Management**

- Blood collection
- Blood stock update
- Blood issue tracking
- Blood group management

**3. Expiration Monitoring**

- Monitor expiry dates
- Generate alerts
- Remove expired units

**4. AI Demand Prediction**

- Analyze historical demand
- Predict future requirements
- Recommend stock levels

**5. Emergency Donor Finder**

- Search nearby donors
- Check compatibility
- Verify eligibility
- Send emergency notifications

**6. Notification System**

- SMS alerts
- Email notifications
- Push notifications

**7. IoT Monitoring**

- Temperature monitoring
- Storage condition monitoring
- Refrigerator status alerts

**Functional Requirements**

- User registration and login
- Blood inventory management
- Donor database management
- Blood request processing
- AI prediction
- Blood expiration tracking
- Emergency donor search
- Alert generation
- Report generation

**Non-Functional Requirements**

**Performance**

- Fast search results
- Real-time updates

**Security**

- Data encryption
- Secure login
- Role-based access

**Reliability**

- 24×7 availability
- Automatic backup

**Scalability**

- Support multiple hospitals
- Handle large donor databases

**Maintainability**

- Modular software architecture
- Easy updates

**Usability**

- User-friendly interface
- Mobile support

**Software Engineering Principles Used**

| **Principle** | **Implementation** |
|---|---|
| Modularity | Separate modules for inventory, donors, AI, alerts |
| Scalability | Cloud infrastructure supports multiple blood banks |
| Maintainability | Independent modules simplify updates |
| Reliability | Automatic backups and monitoring |
| Usability | Simple dashboards and mobile applications |
| Security | Authentication, encryption, access control |

**5. Justification of SDLC Methodology**

**Selected Methodology**

**Agile (Scrum)**

**Why Agile?**

The requirements of healthcare software may change due to government regulations, user feedback, and technological improvements. Agile allows continuous development and testing with frequent user involvement.

Advantages include:

- Faster delivery
- Continuous improvement
- Better collaboration
- Easy requirement changes
- Early bug detection
- Frequent testing
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**Scrum Process**

1. Product Backlog
2. Sprint Planning
3. Sprint Development
4. Testing
5. Review
6. Deployment
7. Maintenance

**How Agile Supports the Project**

- Continuous customer feedback
- Regular software updates
- Easier maintenance
- Better quality assurance
- Reduced project risks
- Faster implementation of new features

## 6. Expected Outcomes and Limitations

**Expected Benefits**

- Better blood availability
- Reduced blood wastage
- Faster donor identification
- Improved emergency response
- Accurate demand forecasting
- Better inventory management
- Secure donor records
- Increased donor participation

**Addressing the Challenges**

| Challenge | Proposed Solution |
|---|---|
| Blood shortage | AI demand prediction |
| Expired blood | Expiration monitoring |
| Slow donor search | GPS-based donor finder |
| Manual inventory | Digital inventory management |
| Emergency delays | Automated alerts |
| Poor donor participation | Reminder notifications |

**Risks and Limitations**

- Initial implementation cost
- Dependence on internet connectivity
- Data privacy and cybersecurity concerns
- Need for staff training
- IoT hardware maintenance
- AI prediction accuracy depends on historical data quality

**Future Enhancements**

- Blockchain for secure blood record management

- Integration with national healthcare systems
- AI chatbot for donor support
- QR code-based blood bag tracking
- Machine learning-based donor recommendation
- Drone delivery of blood in emergencies
- Wearable health integration for donor eligibility
- Multi-language mobile application
- Predictive analytics dashboard for healthcare authorities

## Conclusion

The Intelligent Blood Bank Inventory and Donor Management Platform provides a comprehensive digital solution to modernize blood bank operations. By integrating AI, IoT, cloud computing, GPS-based donor search, and predictive analytics, the system improves inventory management, reduces blood wastage, enhances emergency response, and ensures timely availability of safe blood. The use of the Agile development methodology enables continuous improvement, flexibility, and maintainability, making the platform scalable and suitable for real-world healthcare environments.